

EE 316

Section 3

Lab 01

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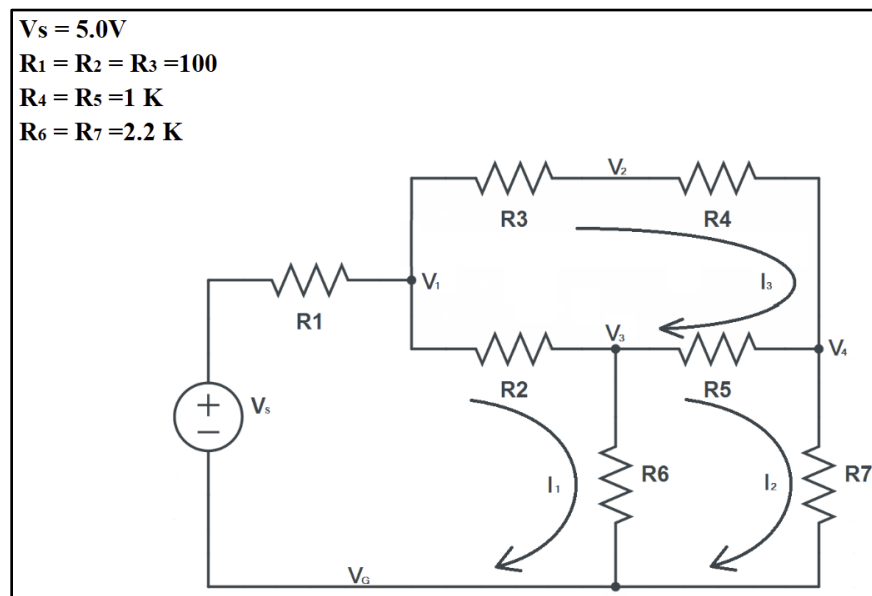
Circuits Review

## Introduction:

The purpose of this lab is to review concepts used to analyze basic circuits such as mesh current analysis, nodal analysis, and Thevenin and Norton equivalent circuits while introducing lab tools such as LT Spice and the lab breadboarding and measurement tools. This lab also serves as a hands-on way to validate basic circuit laws such as Ohm's Law and Kirchoff's Current and Voltage Laws.

## Theoretical Analysis:

### Part 1A



**Figure #1: Circuit Schematic from Lab Manual (Part A)**

Based on Figure 1, mesh analysis (as seen in Figure 2) was used to obtain the following general equations.

$$I_1(R_1 + R_2 + R_6) - I_2R_6 - I_3R_2 = 5V \quad (1)$$

$$-I_1R_6 + I_2(R_5 + R_6 + R_7) - I_3R_5 = 0V \quad (2)$$

$$-I_1R_2 - I_2R_5 + I_3(R_2 + R_3 + R_4 + R_5) = 0V \quad (3)$$

$$\begin{aligned}
I_1: V_S &= I_1 R_1 + R_2 (I_1 - I_3) + R_6 (I_1 - I_2) \\
SV &= I_1 R_1 + I_1 R_2 - I_3 R_2 + I_1 R_6 - I_2 R_6 \\
SV &= I_1 (R_1 + R_2 + R_6) + I_2 (-R_6) + I_3 (-R_2) \\
I_2: 0 &= R_6 (I_2 - I_1) + R_5 (I_2 - I_3) + I_2 R_7 \\
0 &= I_1 (-R_6) + I_2 (R_6 + R_5 + R_7) + I_3 (-R_5) \\
I_3: 0 &= I_3 R_3 + I_3 R_4 + R_5 (I_3 - I_2) + R_2 (I_3 - I_1) \\
0 &= I_1 (-R_2) + I_2 (-R_5) + I_3 (R_3 + R_4 + R_5 + R_2)
\end{aligned}$$

**Figure #2: Mesh Analysis for Part A**

Equations 1-3 were then used to find the mesh currents, branch currents, node voltages, and branch currents for all elements, branches, and meshes in Figure 1. This process was repeated three times with resistor values at 90%, 100%, and 110% of the given values as seen in Table 1.

| Resistor Values | Resistor                                         | Value (Ohms) |
|-----------------|--------------------------------------------------|--------------|
| 90%             | R <sub>1</sub> , R <sub>2</sub> , R <sub>3</sub> | 90           |
|                 | R <sub>4</sub> , R <sub>5</sub>                  | 900          |
|                 | R <sub>6</sub> , R <sub>7</sub>                  | 1980         |
| 100%            | R <sub>1</sub> , R <sub>2</sub> , R <sub>3</sub> | 100          |
|                 | R <sub>4</sub> , R <sub>5</sub>                  | 1000         |
|                 | R <sub>6</sub> , R <sub>7</sub>                  | 2200         |
| 110%            | R <sub>1</sub> , R <sub>2</sub> , R <sub>3</sub> | 110          |
|                 | R <sub>4</sub> , R <sub>5</sub>                  | 1100         |
|                 | R <sub>6</sub> , R <sub>7</sub>                  | 2420         |

**Table #1: Resistor Values for Part 1A**

Using these resistor values, the mesh current values were found using matrix multiplication and equation 4

$$\mathbf{m} = \mathbf{R}^{-1}\mathbf{V} \quad (4)$$

Where  $\mathbf{m}$  is the matrix containing the mesh current values,  $\mathbf{R}^{-1}$  is the inverse of the resistance matrix, and  $\mathbf{V}$  is the matrix containing the voltage source values. Using the mesh current and

resistor values, the node voltages, branch currents, and branch voltages were calculated using equations 5- 8, 9- 15, and Ohm's Law ( $V_n = I_n R_n$ ), respectively.

#### Node Voltages

$$V_1 = V_S - V_{R1} \quad (5)$$

$$V_2 = V_1 - V_{R3} \quad (6)$$

$$V_3 = V_{R6} \quad (7)$$

$$V_4 = V_{R7} \quad (8)$$

#### Branch Currents

$$i_{R1} = I_1 \quad (9)$$

$$i_{R2} = I_1 - I_3 \quad (10)$$

$$i_{R3} = I_3 \quad (11)$$

$$i_{R4} = I_3 \quad (12)$$

$$i_{R5} = I_3 - I_2 \quad (13)$$

$$i_{R6} = I_1 - I_2 \quad (14)$$

$$i_{R7} = I_2 \quad (15)$$

The mesh current, branch current, node voltage, and branch current values are summarized in

Tables 2-4.

| Branch Voltages              | Branch Currents             | Node Voltages           | Loop Currents            |
|------------------------------|-----------------------------|-------------------------|--------------------------|
| $V_{R1} = 363.42 \text{ mV}$ | $i_{R1} = 4.038 \text{ mA}$ | $V_1 = 4.637 \text{ V}$ | $I_1 = 4.038 \text{ mA}$ |
| $V_{R2} = 271.89 \text{ mV}$ | $i_{R2} = 3.021 \text{ mA}$ | $V_2 = 4.544 \text{ V}$ | $I_2 = 1.833 \text{ mA}$ |
| $V_{R3} = 91.53 \text{ mV}$  | $i_{R3} = 1.017 \text{ mA}$ | $V_3 = 4.366 \text{ V}$ | $I_3 = 1.017 \text{ mA}$ |
| $V_{R4} = 915.3 \text{ mV}$  | $i_{R4} = 1.017 \text{ mA}$ | $V_4 = 3.629 \text{ V}$ |                          |
| $V_{R5} = 734.4 \text{ mV}$  | $i_{R5} = 0.816 \text{ mA}$ |                         |                          |
| $V_{R6} = 4.366 \text{ V}$   | $i_{R6} = 2.205 \text{ mA}$ |                         |                          |
| $V_{R7} = 3.629 \text{ V}$   | $i_{R7} = 1.833 \text{ mA}$ |                         |                          |

**Table #2: Theoretical Data for Part A (90% Resistor Values)**

| Branch Voltages             | Branch Currents             | Node Voltages           | Loop Currents            |
|-----------------------------|-----------------------------|-------------------------|--------------------------|
| $V_{R1} = 403.8 \text{ mV}$ | $i_{R1} = 3.634 \text{ mA}$ | $V_1 = 4.637 \text{ V}$ | $I_1 = 3.634 \text{ mA}$ |
| $V_{R2} = 302.2 \text{ mV}$ | $i_{R2} = 2.719 \text{ mA}$ | $V_2 = 4.545 \text{ V}$ | $I_2 = 1.650 \text{ mA}$ |
| $V_{R3} = 101.7 \text{ mV}$ | $i_{R3} = 0.915 \text{ mA}$ | $V_3 = 4.365 \text{ V}$ | $I_3 = 0.915 \text{ mA}$ |
| $V_{R4} = 1.017 \text{ V}$  | $i_{R4} = 0.915 \text{ mA}$ | $V_4 = 3.630 \text{ V}$ |                          |
| $V_{R5} = 0.816 \text{ V}$  | $i_{R5} = 0.735 \text{ mA}$ |                         |                          |
| $V_{R6} = 4.366 \text{ V}$  | $i_{R6} = 1.984 \text{ mA}$ |                         |                          |
| $V_{R7} = 3.629 \text{ V}$  | $i_{R7} = 1.650 \text{ mA}$ |                         |                          |

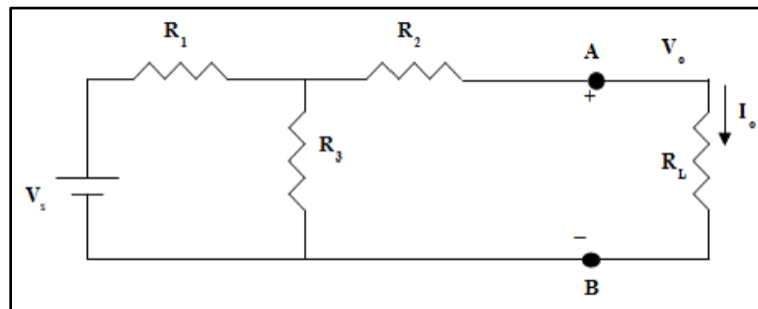
**Table #3: Theoretical Data for Part A (100% Resistor Values)**

| Branch Voltages              | Branch Currents             | Node Voltages           | Loop Currents            |
|------------------------------|-----------------------------|-------------------------|--------------------------|
| $V_{R1} = 363.44 \text{ mV}$ | $i_{R1} = 3.304 \text{ mA}$ | $V_1 = 4.637 \text{ V}$ | $I_1 = 3.304 \text{ mA}$ |
| $V_{R2} = 271.92 \text{ mV}$ | $i_{R2} = 2.472 \text{ mA}$ | $V_2 = 4.545 \text{ V}$ | $I_2 = 1.500 \text{ mA}$ |
| $V_{R3} = 91.52 \text{ mV}$  | $i_{R3} = 0.832 \text{ mA}$ | $V_3 = 4.366 \text{ V}$ | $I_3 = 0.832 \text{ mA}$ |
| $V_{R4} = 1.025 \text{ V}$   | $i_{R4} = 0.932 \text{ mA}$ | $V_4 = 3.630 \text{ V}$ |                          |
| $V_{R5} = 734.8 \text{ mV}$  | $i_{R5} = 0.668 \text{ mA}$ |                         |                          |
| $V_{R6} = 4.366 \text{ V}$   | $i_{R6} = 1.804 \text{ mA}$ |                         |                          |
| $V_{R7} = 3.63 \text{ V}$    | $i_{R7} = 1.500 \text{ mA}$ |                         |                          |

**Table #4: Theoretical Data for Part A (110% Resistor Values)**

## Part 2

Using Figure 3, the open circuit voltage  $V_{OC}$  and short circuit current  $I_{SC}$  across terminals A and B were calculated using a voltage divider and mesh current analysis found in Figures 3 and 4, respectively.



**Figure #3: Circuit Schematic from Lab Manual (Part 2)**

$$\begin{aligned}
 I_{R2} &= 0 & V_2 &= V_s \frac{R_3}{R_1 + R_3} & V_{OC} &= V_2 = 2.5 \text{ V} \\
 & & &= \frac{1}{2} V_s & & \\
 & & &= 2.5 \text{ V} & &
 \end{aligned}$$

**Figure #4:  $V_{OC}$  Calculations using Node Voltage Analysis**

$$\begin{aligned}
 I_1: \quad V_s &= I_1 R_1 + R_3 (I_1 - I_2) \\
 5V &= 2k I_1 - 1k I_2 \\
 I_2: \quad 0V &= R_3 (I_2 - I_1) + I_2 R_2 \\
 0V &= -1k I_1 + 2k I_2 \\
 I_{sc} &= I_2
 \end{aligned}$$

**Figure #5:  $I_{SC}$  Calculations using Mesh Current Analysis**

Next, the Thevenin equivalent resistance  $R_{TH}$  and Norton equivalent resistance  $R_N$  was calculated using equation 16.

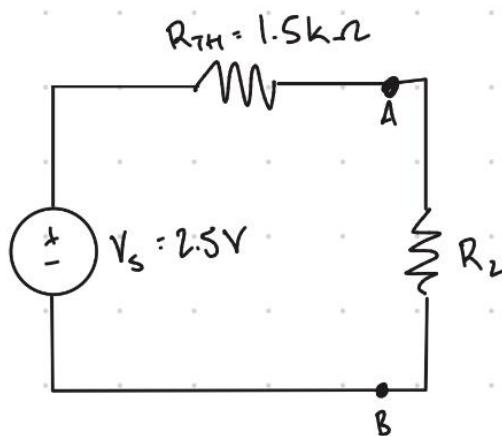
$$R_{TH} = R_N = \frac{V_{OC}}{I_{SC}} \quad (14)$$

A summary of these analytical values— $V_{OC}$ ,  $I_{SC}$ , and  $R_{TH}$ ,  $R_N$ , and  $I_N$ —can be found in Table 5.

|          | Value      |
|----------|------------|
| $V_{OC}$ | 2.5 V      |
| $I_{SC}$ | 1.667 mA   |
| $R_{TH}$ | 4.168 Ohms |
| $R_N$    | 4.168 Ohms |
| $I_N$    | 1.667 mA   |

**Table #5: Thevenin and Norton Analytical Values (Part 1B)**

The Thevenin and Norton equivalent circuits of Figure 3 can be found in Figures 6 and 7.



**Figure #6: Thevenin Equivalent Circuit**

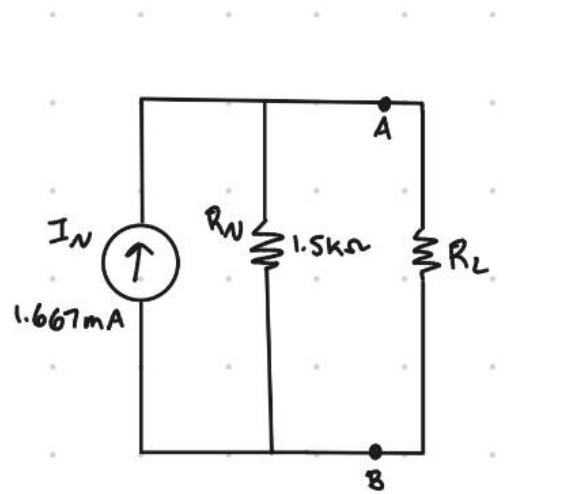


Figure #7: Norton Equivalent Circuit

## Simulation Results:

### Part 1A

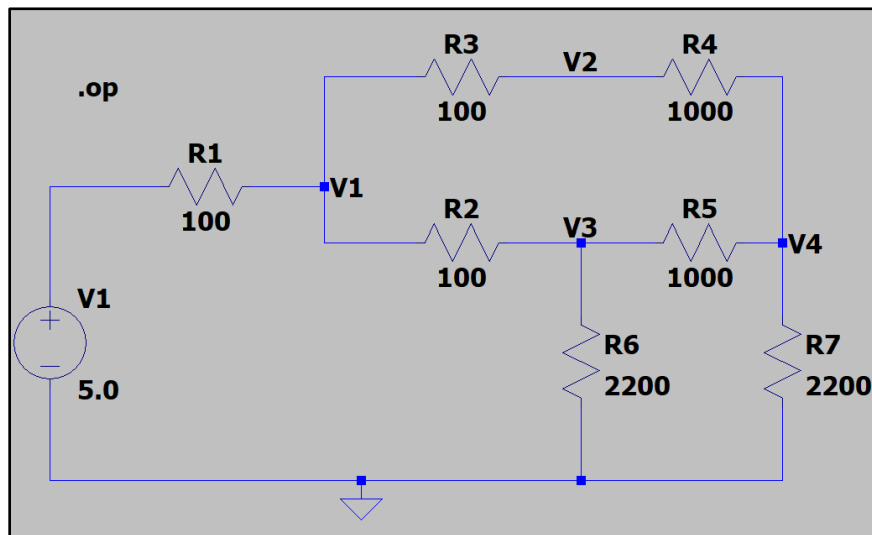


Figure #8: Simulation in LT Spice (Part A)

```

* C:\Users\hanna\OneDrive\Documents\LTspiceXVII\prelab1_part1a.asc

--- Operating Point ---

V(n001):      5          voltage
V(v1):       4.63661    voltage
V(v3):       4.36473    voltage
V(v2):       4.54509    voltage
V(v4):       3.62993    voltage
I(R7):       0.00164997 device_current
I(R6):       0.00198397 device_current
I(R5):       0.000734803 device_current
I(R4):       0.000915164 device_current
I(R3):       0.000915164 device_current
I(R2):       0.00271877 device_current
I(R1):       0.00363393 device_current
I(V1):      -0.00363393 device_current

```

Figure #9: Simulation Output from LT Spice (Part A)

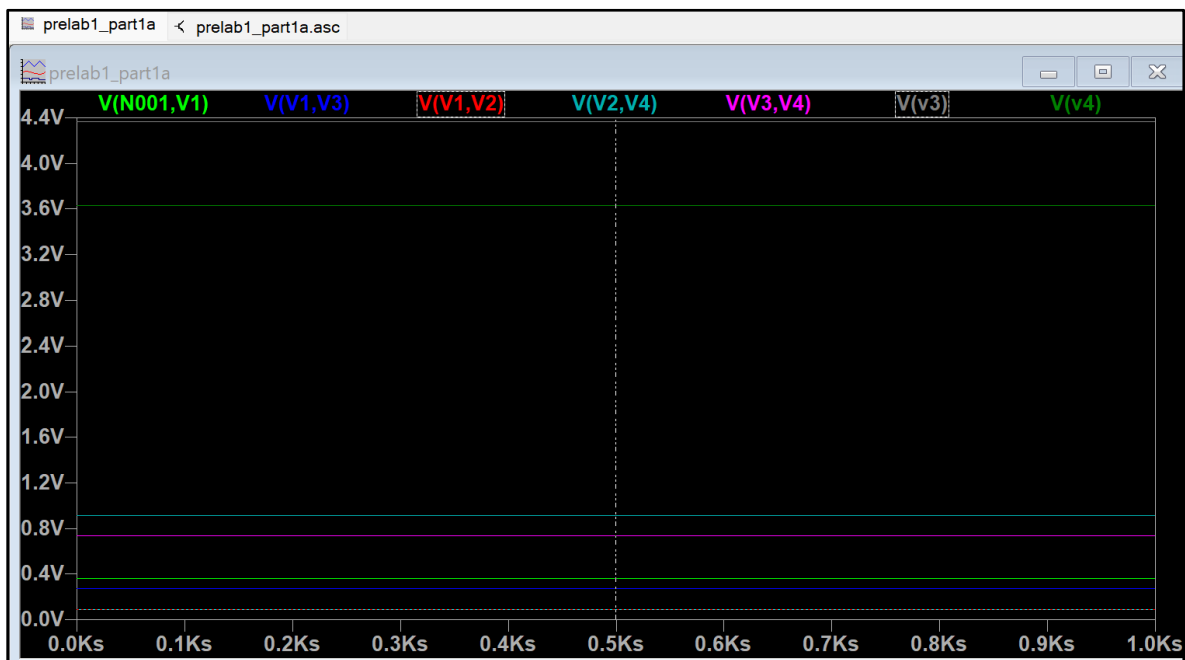


Figure #10: Branch Voltage Data from LT Spice Simulation (Part A)

| Branch Voltages                 | Branch Currents                | Node Voltages             | Loop Currents               |
|---------------------------------|--------------------------------|---------------------------|-----------------------------|
| $V_{R1} = 363.39331 \text{ mV}$ | $i_{R1} = 3.63393 \text{ mA}$  | $V_1 = 4.63661 \text{ V}$ | $I_1 = 3.63393 \text{ mA}$  |
| $V_{R2} = 271.87729 \text{ mV}$ | $i_{R2} = 2.71877 \text{ mA}$  | $V_2 = 4.54509 \text{ V}$ | $I_2 = 1.64997 \text{ mA}$  |
| $V_{R3} = 91.516495 \text{ mV}$ | $i_{R3} = 0.915164 \text{ mA}$ | $V_3 = 4.36473 \text{ V}$ | $I_3 = 0.915164 \text{ mA}$ |
| $V_{R4} = 915.16376 \text{ mV}$ | $i_{R4} = 0.915164 \text{ mA}$ | $V_4 = 3.62993 \text{ V}$ |                             |
| $V_{R5} = 734.80296 \text{ mV}$ | $i_{R5} = 0.734803 \text{ mA}$ |                           |                             |
| $V_{R6} = 4.3647294 \text{ V}$  | $i_{R6} = 1.98397 \text{ mA}$  |                           |                             |
| $V_{R7} = 3.6299264 \text{ V}$  | $i_{R7} = 1.64997 \text{ mA}$  |                           |                             |

Table #6: Simulation Data for Part A

## Part 1B

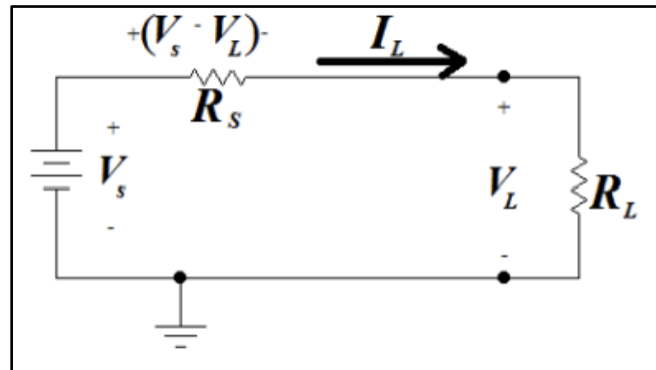


Figure #11: Circuit from Lab Manual (Part B)

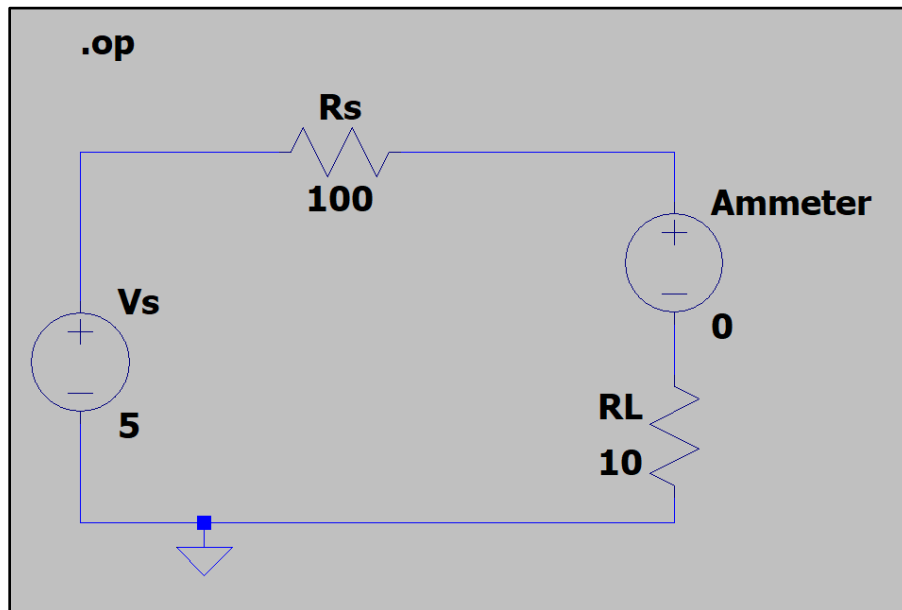


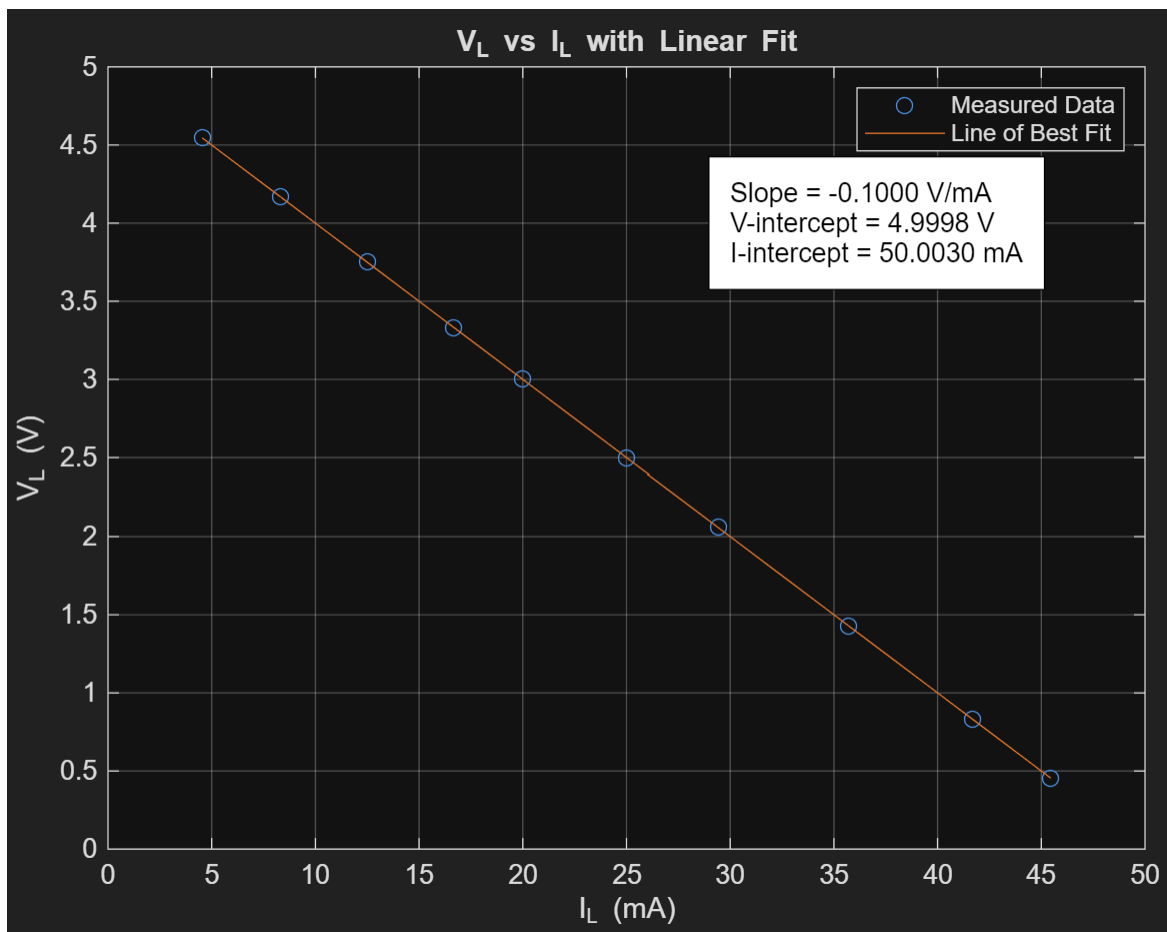
Figure #12: Simulation in LT Spice at  $R_L = 10$  Ohms (Part B)

| --- Operating Point --- |            |                |
|-------------------------|------------|----------------|
| V(n002) :               | 0.454545   | voltage        |
| V(n001) :               | 5          | voltage        |
| V(n003) :               | 0.454545   | voltage        |
| I (RL) :                | -0.0454545 | device_current |
| I (Rs) :                | -0.0454545 | device_current |
| I (Ammeter) :           | 0.0454545  | device_current |
| I (Vs) :                | -0.0454545 | device_current |

Figure #13: Output of Part B Simulation at  $R_L = 10$  Ohms

| $R_L$ (Ohms) | $V_L$ (V)    | $I_L$ (mA)   |
|--------------|--------------|--------------|
| 10           | 454.54547 mV | 45.454547 mA |
| 20           | 833.33331 mV | 41.666668 mA |
| 40           | 1.4285715 V  | 35.714287mA  |
| 70           | 2.0588236 V  | 29.411765mA  |
| 100          | 2.5 V        | 25 mA        |
| 150          | 3 V          | 20 mA        |
| 200          | 3.3333333 V  | 16.666668 mA |
| 300          | 3.75 V       | 12.5 mA      |
| 500          | 4.1666665 V  | 8.3333338 mA |
| 1000         | 4.5454545 V  | 4.5454544 mA |

**Table #7: Simulation Data for Part B**



**Figure #14: Load Voltage vs. Load Current (MATLAB)**

```

IL = [45.455 41.667 35.714 29.412 25 20 16.667 12.5 8.333 4.545]; % in Amps
VL = [0.455 0.833 1.429 2.059 2.5 3 3.333 3.75 4.167 4.545]; % in Volts

p = polyfit(IL, VL, 1); % eq for line of best fit
VL_fit = polyval(p, IL);
slope = p(1);
y_intercept = p(2);
x_intercept = -y_intercept / slope;

plot (IL, VL, 'o')
hold on
plot(IL, VL_fit)
hold off

xlabel('I_L (mA)');
ylabel('V_L (V)');
title('V_L vs I_L with Linear Fit');
legend('Measured Data', 'Line of Best Fit', 'Location', 'northeast');
grid on;

text(30, 4.0, 11print(['Slope = %.4f V/mA\n' ...
    'V-intercept = %.4f V\n' ...
    'I-intercept = %.4f mA'], ...
    slope, y_intercept, x_intercept), ...
    'Color', 'black', ...
    'BackgroundColor', 'white', ...
    'EdgeColor', 'black', ...
    'Margin', 8);

```

**Table #8: MATLAB Code Used to Produce Figure 8**

## Part 2

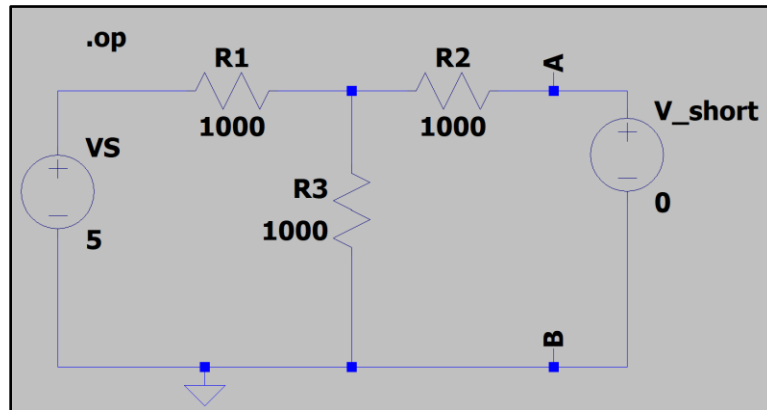


Figure #15: Simulated Circuit for Part 2 (V\_short)

\* C:\Users\hanna\OneDrive\Documents\LTspiceXVII\part2\_vshort.asc

| --- Operating Point --- |             |                |
|-------------------------|-------------|----------------|
| V(n002) :               | 1.66667     | voltage        |
| V(n001) :               | 5           | voltage        |
| V(a) :                  | 0           | voltage        |
| I(R3) :                 | -0.00166667 | device_current |
| I(R2) :                 | -0.00166667 | device_current |
| I(R1) :                 | -0.00333333 | device_current |
| I(V_short) :            | 0.00166667  | device_current |
| I(Vs) :                 | -0.00333333 | device_current |

Figure #16: Simulated Outputs of V\_short

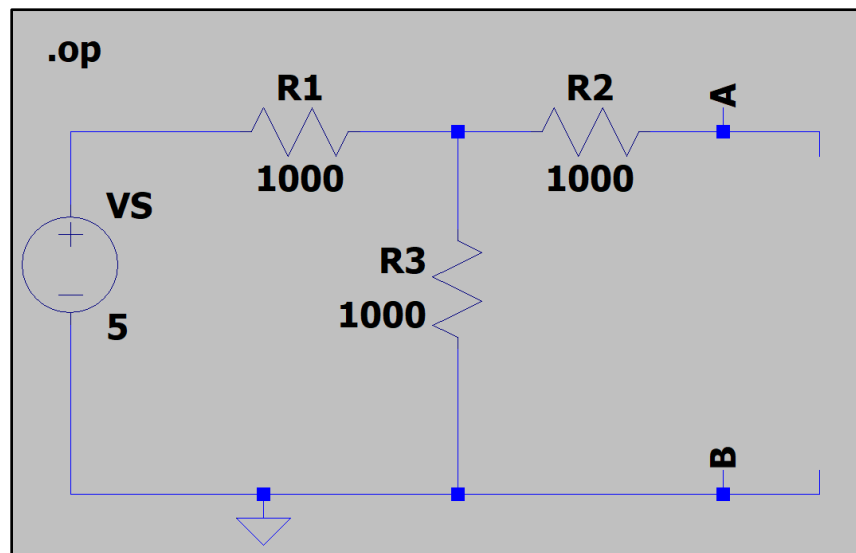


Figure #17: Simulated Circuit for Part 2 (I\_oc)

|                                                                  |         |                |
|------------------------------------------------------------------|---------|----------------|
| * C:\Users\hanna\OneDrive\Documents\LTspiceXVII\part2_vshort.asc |         |                |
| --- Operating Point ---                                          |         |                |
| V(n002) :                                                        | 2.5     | voltage        |
| V(n001) :                                                        | 5       | voltage        |
| V(a) :                                                           | 2.5     | voltage        |
| I(R3) :                                                          | -0.0025 | device_current |
| I(R2) :                                                          | 0       | device_current |
| I(R1) :                                                          | -0.0025 | device_current |
| I(Vs) :                                                          | -0.0025 | device_current |

Figure #18: Simulated Outputs of  $I_{oc}$

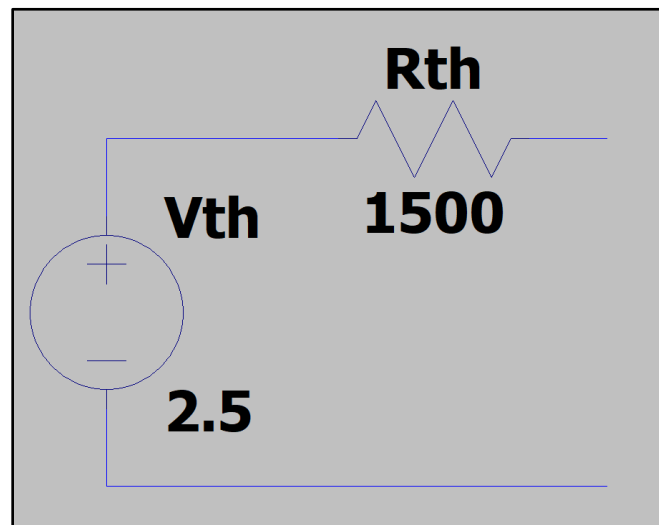


Figure #19: Thevenin Equivalent Circuit

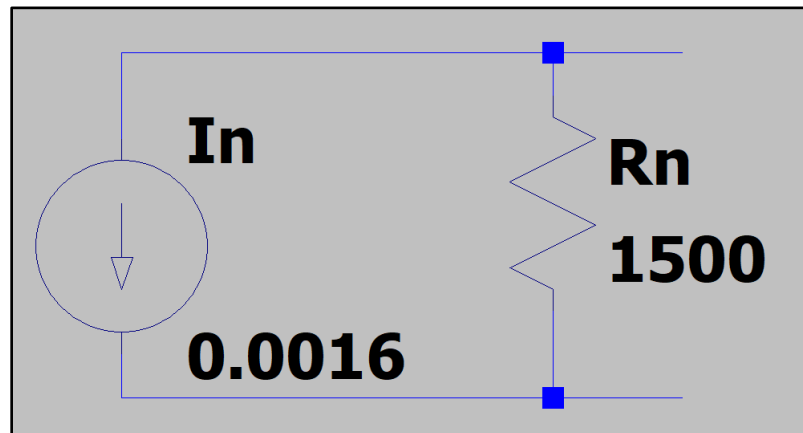


Figure #20: Norton Equivalent Circuit

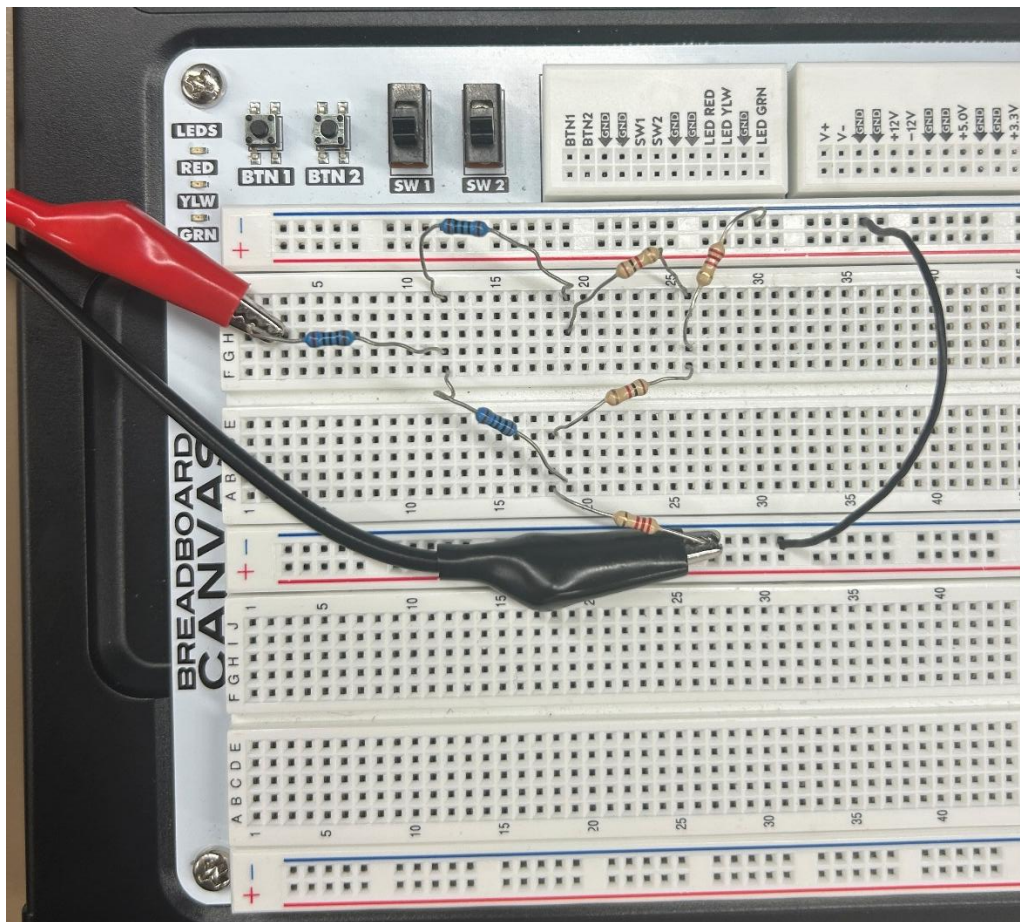
| Circuit  | $R_L$ (Ohms) | $P_{max}$                                                            |
|----------|--------------|----------------------------------------------------------------------|
| Thevenin | 1500         | $P_{max} = \frac{V_{th}^2 R_L}{2R_{TH}} = 1.042 \text{ mW}$          |
| Norton   | 1500         | $P_{max} = \frac{I_N^2 R_N^2 R_L}{(R_N + R_L)^2} = 0.960 \text{ mW}$ |

**Table #9: Power Calculations of Thevenin and Norton Circuits**

## Experimental Setup and Results:

### Part 1A

The circuit diagram found in Figure 1 was built using the Analog/Digital Trainer in the electronic laboratory. A picture of the breadboarded circuit can be found in Figure 21. Using this circuit and a digital multimeter, the branch voltages, branch currents, node voltages, and loop currents were measured and recorded in Table 10.



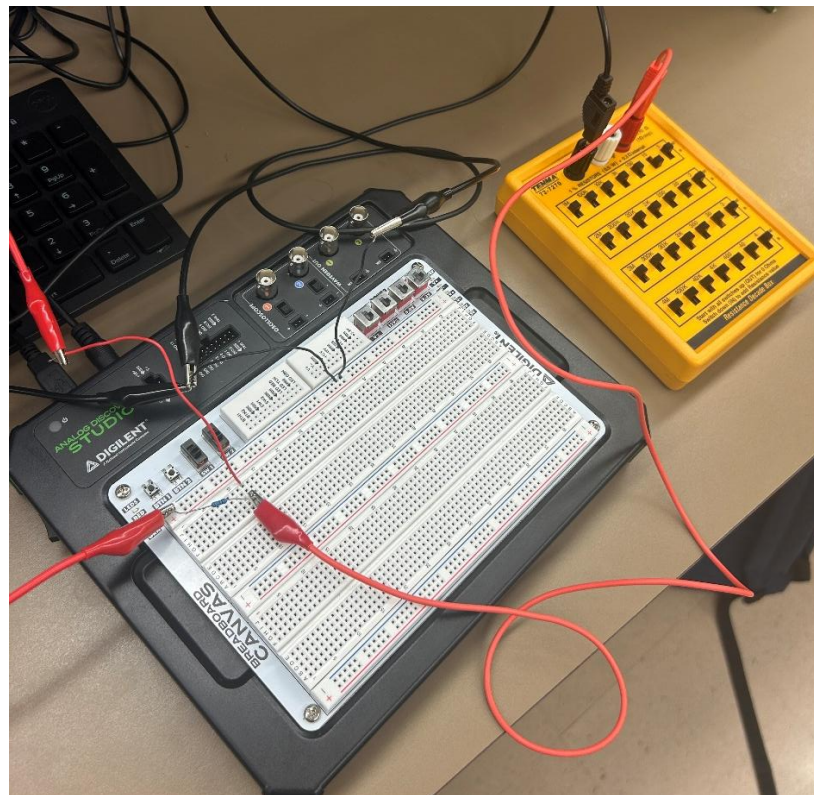
**Figure #21: Experimental Setup (Part 1A)**

| Branch Voltages             | Branch Currents             | Node Voltages           | Loop Currents            |
|-----------------------------|-----------------------------|-------------------------|--------------------------|
| $V_{R1} = 368.5 \text{ mV}$ | $i_{R1} = 3.44 \text{ mA}$  | $V_1 = 4.632 \text{ V}$ | $I_1 = 3.44 \text{ mA}$  |
| $V_{R2} = 276.8 \text{ mV}$ | $i_{R2} = 2.70 \text{ mA}$  | $V_2 = 4.538 \text{ V}$ | $I_2 = 1.669 \text{ mA}$ |
| $V_{R3} = 93.6 \text{ mV}$  | $i_{R3} = 0.8 \text{ mA}$   | $V_3 = 4.408 \text{ V}$ | $I_3 = 0.924 \text{ mA}$ |
| $V_{R4} = 925.4 \text{ mV}$ | $i_{R4} = 0.924 \text{ mA}$ | $V_4 = 3.666 \text{ V}$ |                          |
| $V_{R5} = 742.3 \text{ mV}$ | $i_{R5} = 0.737 \text{ mA}$ |                         |                          |
| $V_{R6} = 4.408 \text{ V}$  | $i_{R6} = 2.00 \text{ mA}$  |                         |                          |
| $V_{R7} = 3.666 \text{ V}$  | $i_{R7} = 1.669 \text{ mA}$ |                         |                          |

**Table #10: Experimental Results (Part A)**

## Part 1B

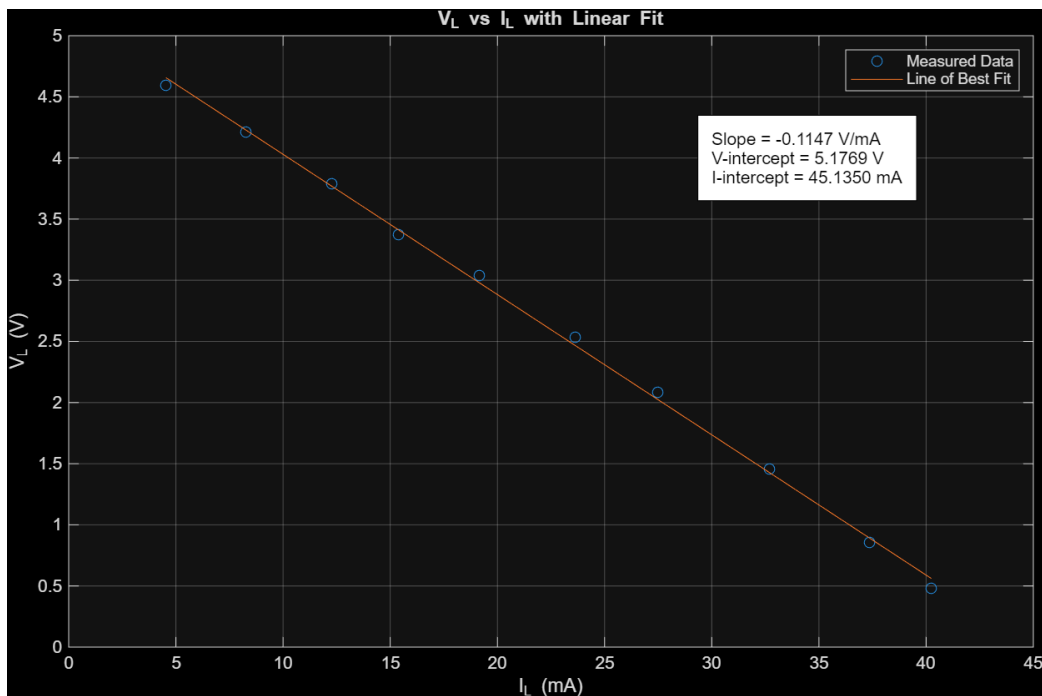
The circuit in Figure 11 was built using a variable resistor. A picture of the experimental setup can be found in Figure 22. The current and voltage across the load resistor was then measured for each resistor value and recorded in Table 11. The results were then plotted in MATLAB and the line of best fit, x-intercept, and y-intercept were obtained using the code found in Table 8. This plot can be found in Figure 23.



**Figure #22: Experimental Setup with Variable Resistor (Part 1B)**

| $R_L$ (Ohms) | $V_L$ (V) | $I_L$ (mA) |
|--------------|-----------|------------|
| 10           | 482.1 mV  | 40.25 mA   |
| 20           | 859.1 mV  | 37.349 mA  |
| 40           | 1.454 V   | 32.687 mA  |
| 70           | 2.087 V   | 27.490 mA  |
| 100          | 2.533 V   | 23.612 mA  |
| 150          | 3.036 V   | 19.152 mA  |
| 200          | 3.373 V   | 15.393 mA  |
| 300          | 3.789 V   | 12.277 mA  |
| 500          | 4.211 V   | 8.262 mA   |
| 1000         | 4.595 V   | 4.539 mA   |

**Table #11: Experimental Results (Part 1B)**



**Figure #23: MATLAB Plot of Experimental Results (Part 1B)**

## Comparison and Discussion:

### Part 1A

The experimental results found in Table 6 differ from the analytical results found in Tables 2-4, however, the experimental results have a percent error of about 1-2%, depending on the specific measurement. This relatively small difference indicates that the circuit components are operating within their device tolerances, and the experimental voltage and current values fall within the

expected operational ranges. The close agreement between the analytical and experimental results also demonstrates that the analytical analysis accurately represents the behavior of the circuit. The differences between the analytical and experimental results can be attributed to several factors, including component tolerances, measurement inaccuracies, and variations in the voltage supplied to the circuit compared to its nominal value. Overall, the small percent errors indicate that the experimental results are consistent with the theoretical analysis and that the circuit performed as expected.

## **Part 1B**

The experimental results found in Table 11 differ from the analytical results in Table 7. The experimental results have percent errors ranging from 1-3.1%, again indicating that the devices are operating within operational ranges. While the experimental results vary from the analytical ones, they are within a normal range. All the components in the circuit are operating correctly.

Additionally, the graphs of  $V_L$  vs.  $I_L$  for the analytical and experimental results found in Figures 14 and 23 are remarkably similar. The slope has a percent error of 14.7% while the voltage and current intercepts have percent errors of 3.54% and 9.7% respectively. While this circuit has higher percent errors overall when compared to Part 1, the circuit is still operating within expected ranges. Like in the first circuit, the discrepancies between the analytical and experimental results can be a result of many factors including imperfect components, measurement inaccuracies, and an imperfect voltage source.

## **Conclusion:**

This lab provided an opportunity to apply several fundamental circuit analysis techniques, including mesh current analysis, nodal analysis, and Thevenin and Norton Equivalent circuits. The circuits were analyzed using analytical methods, simulated using LT Spice, and constructed

and measured using a digital multimeter. This allowed the theoretical concepts discussed in class to be applied to physical circuits and provided practical experience with the tools used to analyze and measure electrical systems.

The results of this lab demonstrated the usefulness of these circuit analysis techniques for determining circuit behavior and simplifying complex circuits. Additionally, the lab reinforced fundamental electrical principles such as Ohm's Law and Kirchhoff's Current and Voltage Laws. Overall, the lab connected theoretical circuit analysis with hands-on experimentation, providing a practical understanding of how these methods can be used to analyze real-world circuits.